

DECK INDEX • PART 1

01:

Cell Theory

02:

Mendel's Laws

03:

DNA Double Helix

04:

05:

06:

07:

08:

09:

10:

11:

12:

13:

14:

15:

YAADCARD.IN

CELL
THEORY

YAADCARD.IN

MENDEL'S
LAWS

YAADCARD.IN

DNA
DOUBLE
HELIX

YAADCARD.IN

CENTRAL
DOGMA

YAADCARD.IN

NATURAL
SELECTION

YAADCARD.IN

LAMARCKISM

YAADCARD.IN

MUTATION
THEORY

YAADCARD.IN

OPERON
CONCEPT

YAADCARD.IN

ABO
BLOOD
GROUPS

YAADCARD.IN

FIRST
VACCINATION

YAADCARD.IN

GERM
THEORY

YAADCARD.IN

YAADCARD • BIOLOGY

FLASH CARDS

BIOLOGY

yaadcard.in

PRODUCT DESCRIPTION

Premium high-yield study deck engineered for quick revision and active recall. Contains concise cards covering major core concepts, visual diagrams, mathematical formulas, and memory mnemonics. Perfect for exam preparation, competitive tests, and classroom revision.

Pack Contents: Premium Flashcards & reference insert
Subject: Biology
MRP: ₹250.00 (incl. of all taxes)
Support: support@yaadcard.in

YAADCARD Publications
Made in India • Recyclable

▼ FOLD TOP ▼

▼ FOLD BOTTOM ▼

YAADCARD – PRINT, CUT, FOLD, AND PACK

NATURAL SELECTION

Survival of the fittest

Organisms better adapted to their environment survive and reproduce more successfully.

Charles Darwin, 1859
Origin of Species; survival of fittest

CENTRAL DOGMA

How genetic information flows

Describes the flow of genetic information within a biological system, from gene to protein.

Proposed by Francis Crick (1958)
Reverse transcription is an exception

DNA→RNA→Protein

DOUBLE HELIX

Structure of the DNA molecule

Two antiparallel strands twisted into a helix, held by hydrogen bonds between base pairs.

Watson & Crick, 1953
Based on Rosalind Franklin's X-ray data

MENDEL'S LAWS

Foundation of classical genetics

Gregor Mendel's pea plant experiments established the basic rules of trait inheritance.

Law of Dominance & Segregation
Law of Independent Assortment

CELL THEORY

All life is made of cells

States that cells are the basic unit of life, and all cells arise from pre-existing cells.

Schleiden & Schwann (1838-39)
Virchow: Omnis cellula e cellula

DECK INDEX • PART 2

16: Endosymbiotic Theory
17: 5 Kingdom System
18: Three Domain System
19: Binomial Nomenclature
20: Hardy-Weinberg
21: PCR
22: Restriction Enzymes
23: Human Genome Project
24: Insulin Sequencing
25: First Cloned Mammal
26: Reflex Arc
27: Neuromuscular Junction
28: Blood Sugar Control
29: Menstrual Cycle

YAADCARD.IN

OPERON CONCEPT

How bacterial genes switch on

Explains gene regulation in bacteria; genes are turned on/off together as a unit.

Jacob & Monod, 1961
Lac operon regulates gene expression

MUTATION THEORY

Evolution by sudden change

Proposes that evolution occurs through sudden, large mutations rather than gradual change.

Proposed by Hugo de Vries
Evolution via sudden mutations

LAMARCKISM

Inheritance of acquired traits

Theory that traits acquired during an organism's life can be passed to its offspring.

Jean-Baptiste Lamarck's theory
Acquired characters are inherited

GERM THEORY

Microbes cause disease

Established that many diseases are caused by microorganisms, not spontaneous generation.

Louis Pasteur & Robert Koch
Microbes cause infectious disease

FIRST VACCINE

Birth of immunization

The first successful vaccine used cowpox material to protect against smallpox infection.

Edward Jenner, 1796
First vaccine: smallpox via cowpox

ABO BLOOD GROUPS

Human blood typing system

Classification of blood based on presence or absence of A and B surface antigens.

Karl Landsteiner, 1900
4 types: A, B, AB, O

PHOTOSYNTHESIS

YAADCARD.IN

KREBS
CYCLE

YAADCARD.IN

GLYCOLYSIS

YAADCARD.IN

CHEMIOSMOTIC
THEORY

YAADCARD.IN

ENDOSYMBIOTIC
THEORY

YAADCARD.IN

5 KINGDOM
SYSTEM

YAADCARD.IN

THREE
DOMAIN
SYSTEM

YAADCARD.IN

BINOMIAL
NOMENCLATURE

YAADCARD.IN

HARDY-WEINBERG

YAADCARD.IN

PCR

YAADCARD.IN

RESTRICTION
ENZYMES

YAADCARD.IN

HUMAN
GENOME
PROJECT

YAADCARD.IN

INSULIN
SEQUENCING

YAADCARD.IN

FIRST
CLONED
MAMMAL

YAADCARD.IN

REFLEX
ARC

YAADCARD.IN

NEUROMUSCULAR
JUNCTION

YAADCARD.IN

BLOOD
SUGAR
CONTROL

YAADCARD.IN

MENSTRUAL
CYCLE

YAADCARD.IN

5 KINGDOMS Classifying all living organisms Organisms are grouped into five kingdoms based on cell structure and nutrition mode. R.H. Whittaker, 1969 5 Kingdoms: Monera to Animalia	ENDOSYMBIOSIS Origin of organelles Mitochondria and chloroplasts evolved from free-living bacteria engulfed by ancestral cells. Proposed by Lynn Margulis From engulfed bacteria, origin theory	CHEMIOSMOSIS How mitochondria make ATP Explains ATP synthesis via a proton gradient across the inner mitochondrial membrane. Proposed by Peter Mitchell Explains ATP synthase mechanism	GLYCOLYSIS First step of respiration Breakdown of glucose into pyruvate, producing a small net gain of ATP and NADH. Occurs in the cytoplasm Net gain: 2 ATP, 2 NADH	KREBS CYCLE Citric acid energy cycle Series of reactions that release stored energy from carbohydrates, fats, and proteins. Hans Krebs, 1937 Occurs in mitochondrial matrix	PHOTOSYNTHESIS Light to chemical energy conversion Plants convert carbon dioxide and water into glucose and oxygen using sunlight. $6CO_2 + 6H_2O \rightarrow C_6H_{12}O_6 + 6O_2$ Occurs in chloroplast thylakoids						
GENOME PROJECT Mapping human DNA International effort to sequence and map every gene in the human genome. Completed in the year 2003 Mapped all human genes	RESTRICTION ENZYMES Molecular scissors for DNA Enzymes that cut DNA at specific recognition sequences, key to genetic engineering. Smith, Arber & Nathans Cut DNA at specific sites	PCR Amplifying DNA in a lab A technique to rapidly make millions of copies of a specific DNA segment. Kary Mullis, 1983 Amplifies DNA exponentially in vitro	HARDY-WEINBERG Population genetic equilibrium Describes allele and genotype frequencies in a non-evolving population over generations. Formula: $p^2 + 2pq + q^2 = 1$ Genetic equilibrium in populations	BINOMIAL NAMING Two-part scientific naming Every species is given a standardized two-part Latin name: genus and species. Carl Linnaeus system Genus + species naming format	3 DOMAINS Highest level of classification Groups all life into three domains based on differences in rRNA sequences. Carl Woese, 1990 Archaea, Bacteria, Eukarya domains						
MENSTRUAL CYCLE Monthly reproductive cycle A roughly 28-day cycle of hormonal changes preparing the body for pregnancy. 4 phases, ~28 day cycle Follicular then Ovulation then Luteal	SUGAR REGULATION Two hormones, opposite jobs Pancreatic hormones work antagonistically to keep blood glucose levels stable. <table><tr><th>Hormone</th><th>Effect</th></tr><tr><td>Insulin</td><td>Lowers sugar</td></tr><tr><td>Glucagon</td><td>Raises sugar</td></tr></table>	Hormone	Effect	Insulin	Lowers sugar	Glucagon	Raises sugar	NMJ Where nerves meet muscles Synapse where a motor neuron communicates with a muscle fiber to trigger contraction. Neurotransmitter: Acetylcholine (ACh) Synapse between nerve & muscle	REFLEX ARC Fastest nerve response pathway The neural pathway that mediates a rapid, involuntary reaction to a stimulus. Receptor to effector pathway Bypasses conscious brain control	DOLLY THE SHEEP First mammal cloned First mammal successfully cloned from an adult somatic cell using nuclear transfer. Ian Wilmut, 1996 Dolly the sheep, Scotland	INSULIN SEQUENCED First protein ever sequenced The complete amino acid sequence of insulin was determined, a landmark in biochemistry. Frederick Sanger, 1955 First protein fully sequenced
Hormone	Effect										
Insulin	Lowers sugar										
Glucagon	Raises sugar										